

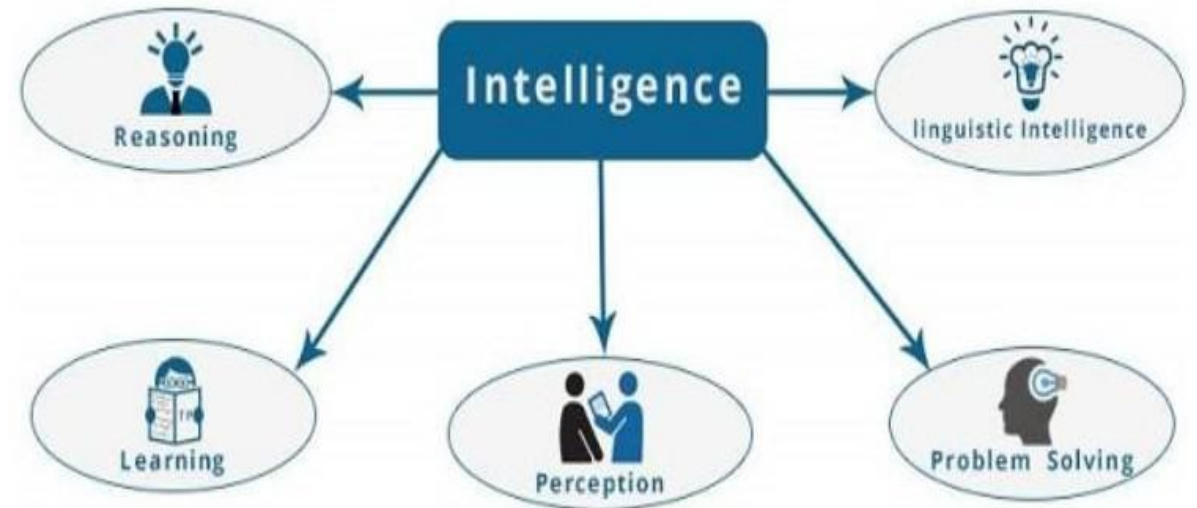
# **LECTURE: 1**

## **ARTIFICIAL INTELLIGENCE ESSENTIALS (INT 428)**

# What is Intelligence?

It is the ability to:

- acquire knowledge and skills,
- solve problems,
- and adapt to new situations.





# Key Aspects of Intelligence:

- Reasoning
- Problem-solving
- Learning
- Perception
- Decision-making

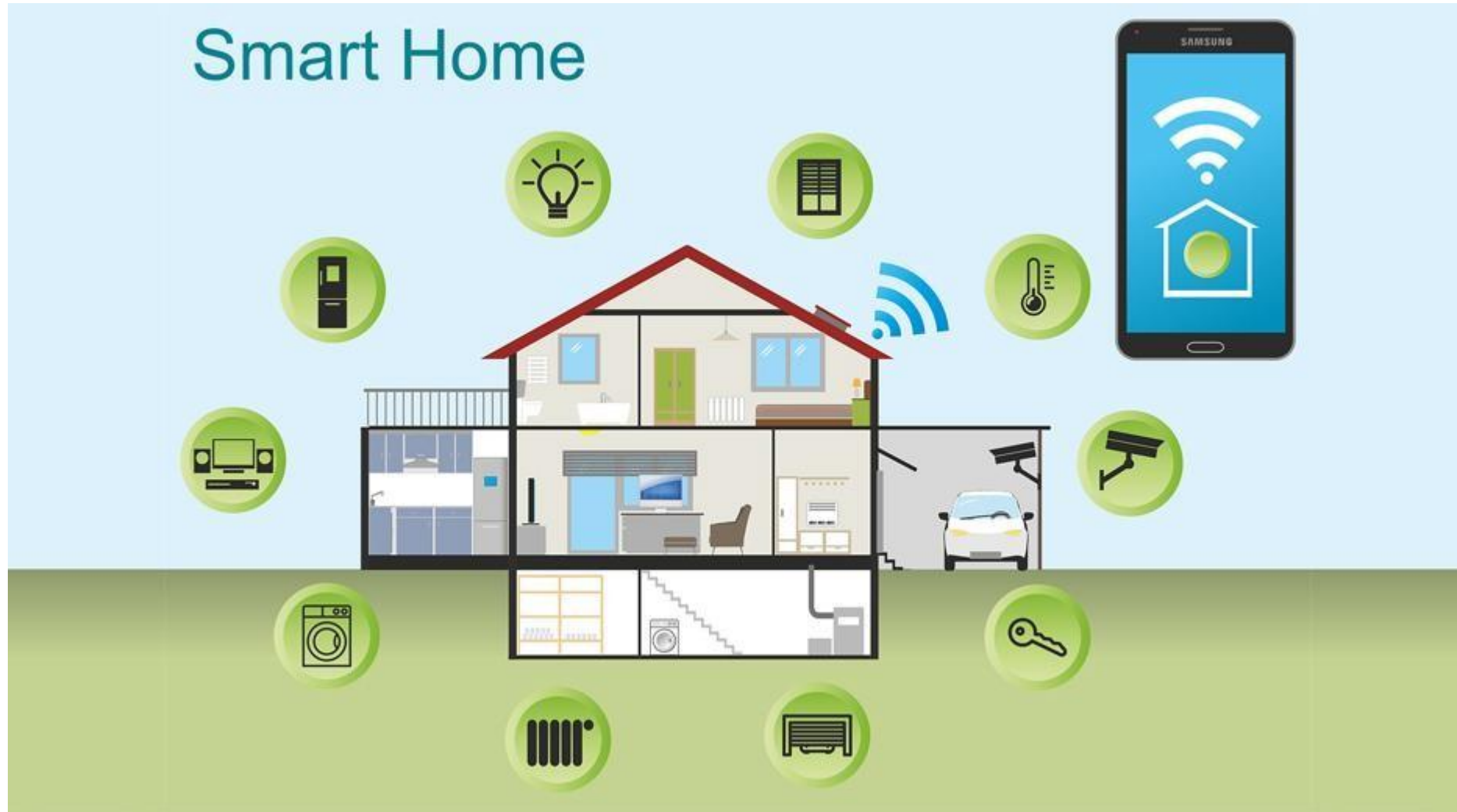


?

What word first comes to mind when you hear the term “Artificial Intelligence?”



# AI Applications: Smart Home

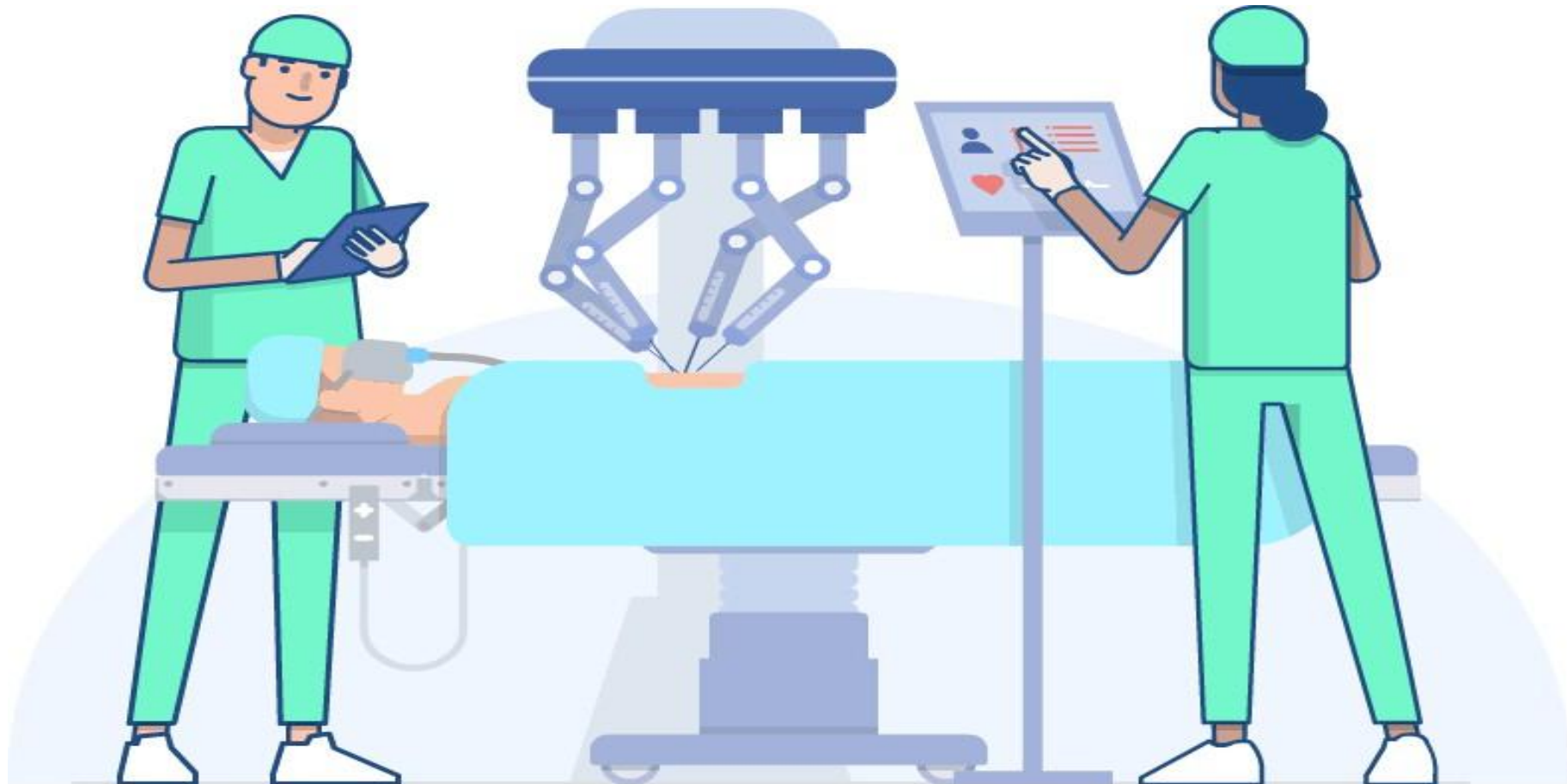


# AI Applications: Smart City





# AI Applications: Medical



# AI Applications: Agriculture





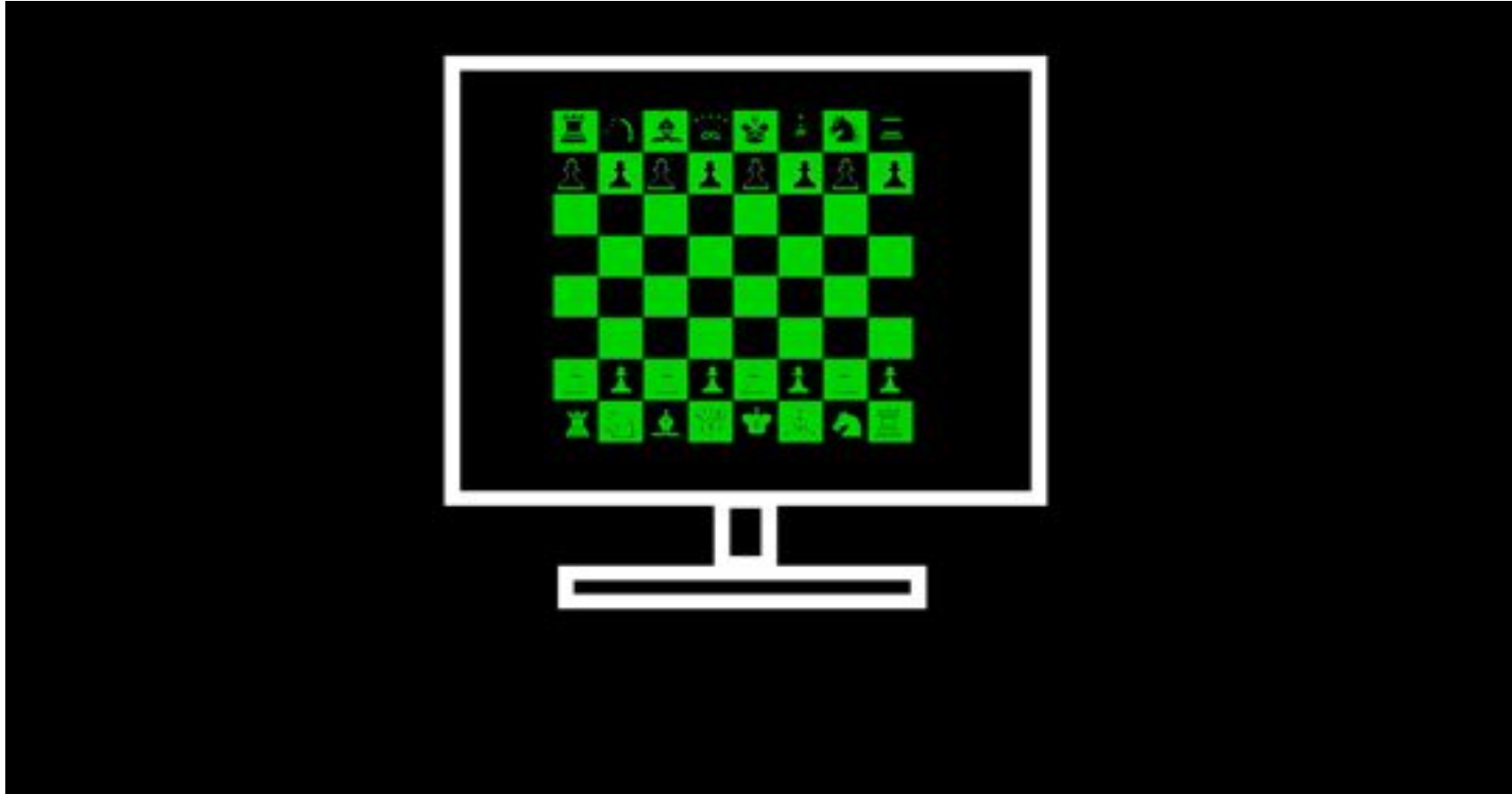
# AI Applications: Stock Market



# AI Applications: Transportation

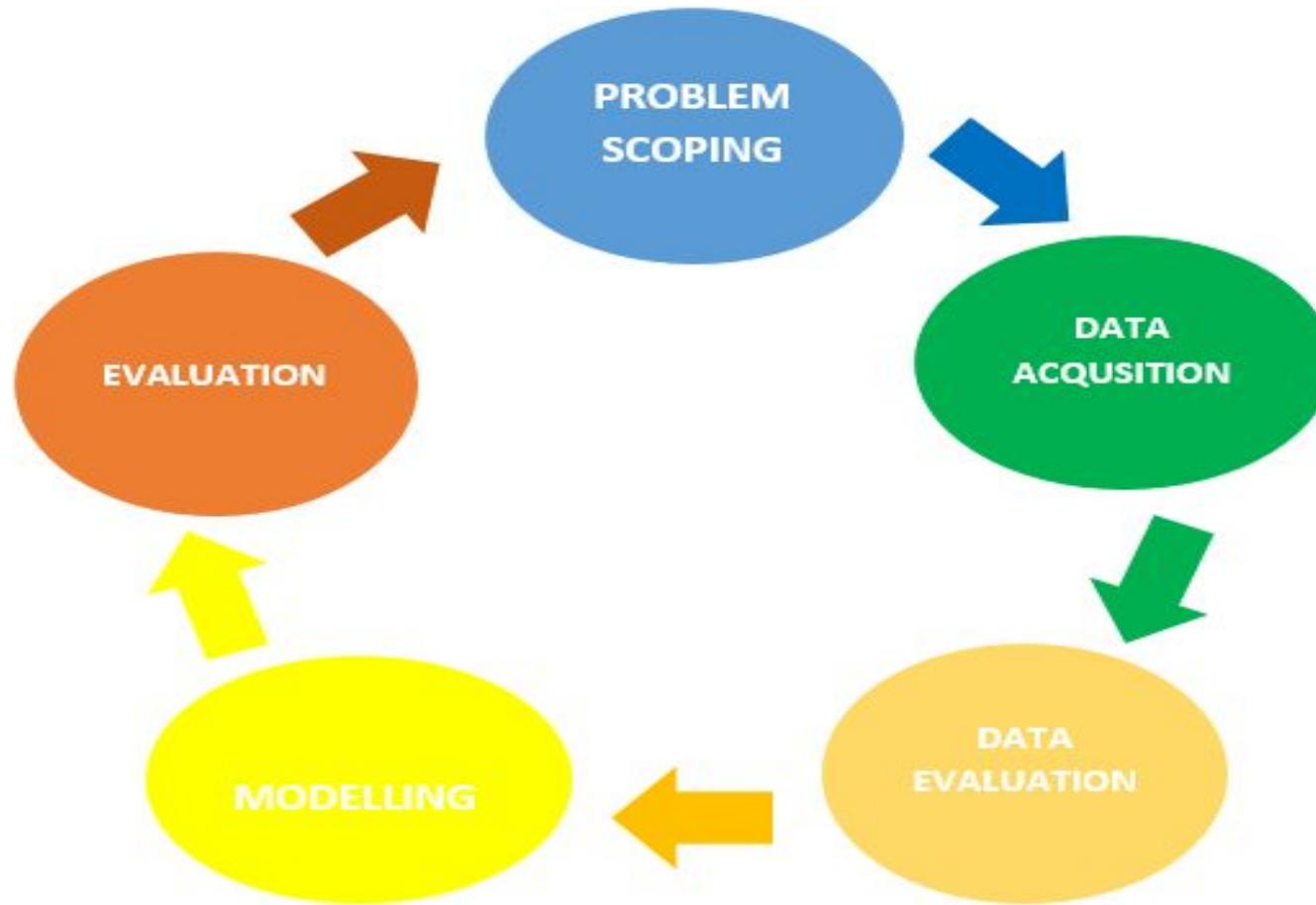


# AI Applications: Gaming





# AI Project Cycle





# What is AI?

|                  |                     |
|------------------|---------------------|
| Thinking humanly | Thinking rationally |
| Acting humanly   | Acting rationally   |

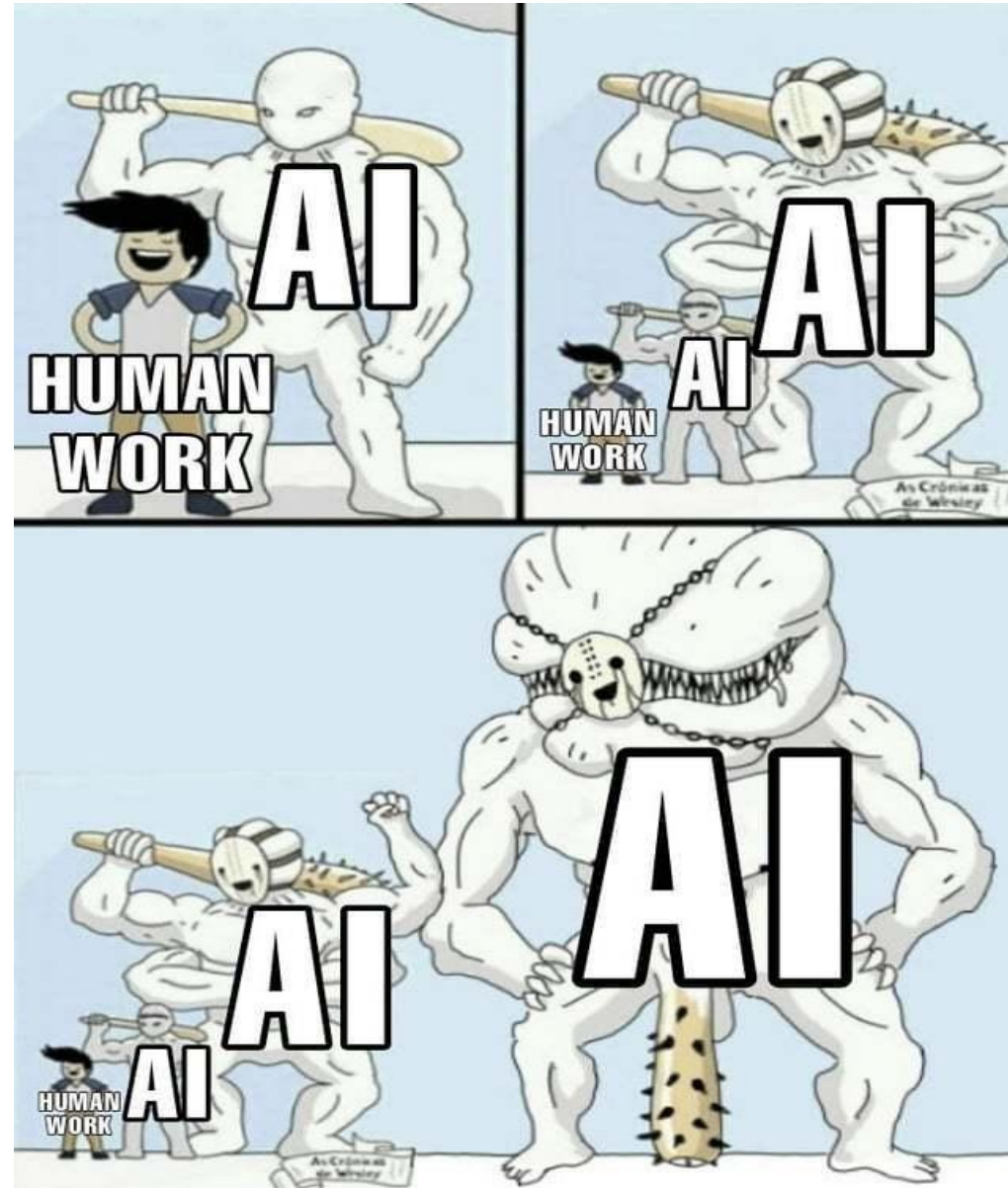


**Human Intelligence:** Trial, error, observation, intuition, adaptation, creativity.

**Artificial Intelligence:** Programming, training, optimizing, automating, simulating, replicating.



**Boston Dynamics'**



# What is Artificial Intelligence?

- AI is a branch of computer science.
- Develops machines that simulate human intelligence.
- Enables machines to think and learn.
- Focuses on solving complex human problems.





# Goals of AI:

- Automate tasks
- Mimic human cognition
- Solve complex problems efficiently





# Foundations of Artificial Intelligence:

- Philosophy (Logic and Reasoning)
- Mathematics (Probability, Statistics, Algorithms)
- Psychology (Human Behavior)
- Computer Engineering (Hardware/Software)
- Neuroscience (Brain Functioning)
- Linguistics (Language Processing)

***NEXT- AI's Journey: Dreams to Reality***

*How does AI perform tasks?*

*What makes AI think like humans?*

***Explore AI's inner workings and secrets!***

